## Question 10

**Q:** An image captured with low resolution fails to detect objects clearly. Explain how sampling and image formation affect object visibility.

## Theoretical Concept: Low Resolution and Visibility

**Sampling** defines spatial resolution. If the sampling rate is too low relative to the size of the object, the object will occupy too few pixels to be distinguishable (violating Nyquist).

Image formation factors (like poor focus or motion blur during capture) further spread the object's light across multiple pixels, completely destroying visibility in low-res setups.

## 💻 OpenCV Python Solution

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```
import cv2  # Import OpenCV for image processing algorithms
import numpy as np                                                # Import Numpy for mat
import matplotlib.pyplot as plt                                   # Import Matplotlib to

# Step 1: Create an image with a clear, readable letter 'A'
img = np.zeros((100, 100), dtype=np.uint8)  # Create a blank black image array fill
cv2.putText(img, 'A', (20, 80), cv2.FONT_HERSHEY_SIMPLEX, 3, 255, 5)  # Render text

# Step 2: Simulate an extremely low sampling rate (sensor with few pixels)
low_res = cv2.resize(img, (10, 10))  # Resize image to new dimensions using interpo

# Step 3: Attempt to resize it back up using interpolation to 'enhance' it
# This proves that lost sampling data cannot simply be interpolated back
upscaled = cv2.resize(low_res, (100, 100), interpolation=cv2.INTER_LINEAR)  # Resiz

# Step 4: Show the permanent loss of visibility
plt.figure(figsize=(8,4))  # Create a new figure window for display
plt.subplot(121), plt.imshow(img, cmap='gray'), plt.title('High Sampling')  # Place
```

```
plt.subplot(122), plt.imshow(upscaled, cmap='gray'), plt.title('Low Sampling (Data
plt.show()  # Show the final plot window to the user
```

## 🎯 Expected Output

You will see a crisp letter 'A', next to a blurry, unrecognizable gray blob. Because the initial sampling grid (10×10) was too small, the true shape of the object was permanently erased.

## 🚀 How to Execute & Submit

**1. Google Colab Execution:**

- Open [Google Colab]() and create a New Notebook.
- Click the 📋 **Copy** button on the code block above.
- Paste it into a Colab cell and press **Shift + Enter** to run.

**2. Uploading to GitHub:**

- In Colab, go to **File** > **Save a copy in GitHub**.
- Authorize your GitHub account.
- Select your Computer Vision Lab repository and click **OK** to commit the code.